

1. What is oops in Python?

* OOPS (Object Oriented Programming System).

It is a programming concept that organizes code using classes and objects. It helps in code reusability, security, and easy maintenance.

Main components of OOPS:

1. Class
2. Object
3. Inheritance
4. Polymorphism
5. Encapsulation
6. Abstraction

2. Components of Python:

1. class - Blueprint for creating objects
2. object - instance of class
3. inheritance - child class inherits properties from parent class
4. polymorphism - same method behaves differently.
5. Encapsulation - Hides data using private members
6. Abstraction - Shows essential details and hides implementation.

3. Access class variable city = "chennai" and pass the same value for all Objects

Class Student:

city = "chennai"

S1 = Student()

S2 = Student()

S3 = Student()

S4 = Student()

S5 = Student()

Print (S1.city)

Print (S2.city)

Print (S3.city)

Print (S4.city)

Print (S5.city)

A. Create a class with instance variables name, age, city and create 5 objects.

Class Student:

def __init__(self, name, age, city):

self.name = name

self.age = age

self.city = city

```
def display(self):
```

```
    print("name:", self.name)
```

```
    print("age:", self.age)
```

```
    print("city:", self.city)
```

```
s1 = Student("selva", 21, "chennai")
```

```
s2 = Student("Ajay", 22, "Madurai")
```

```
s3 = Student("Sheik", 20, "Tenkasi")
```

```
s4 = Student("praveen", 24, "salem")
```

```
s5 = Student("Yogesh", 23, "Namakkal")
```

```
s1.display()
```

```
s2.display()
```

```
s3.display()
```

```
s4.display()
```

```
s5.display()
```

5. Types of inheritance:

* Five types of inheritance in Python

1. Single inheritance

2. Multiple inheritance

3. Multilevel inheritance

4. Hierarchical inheritance

5. Hybrid inheritance

6 Scenario for inheritance?

Example 1: School Management.

Person

|

Student.

A Student inherits name and age from Person.

7. Use of `--init--()`?

* `--init--()` is a constructor in Python.

* it is automatically called when an object is created.

* it is used to:

* initialize object data

* Assign values to instance variables

* Avoid writing separate initialization

methods.